Sustainability as a Software Engineering Problem

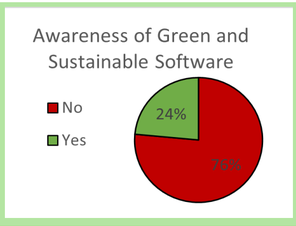
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1. The Problem

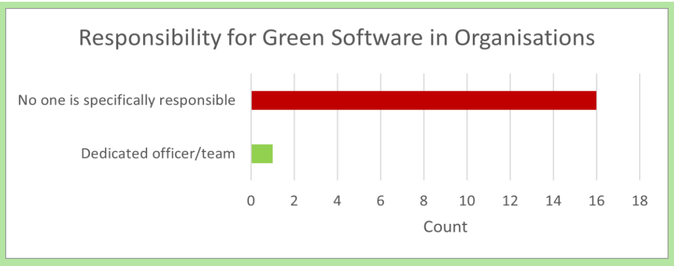
Software is critical to modern life but its sustainability is a rising concern. Running and maintaining software requires significant energy and storage resources, even for unused applications. Many programs and projects are abandoned rather than being reused or retired, adding to digital waste and missing opportunities to apply sustainable practices.

2. Approach

The approach taken was to create an Insights Matrix using data collected from GitHub, mobile application stores and developer surveys. This provides an evidence-based view of current gaps and opportunities for guiding future green software engineering practices.



From our survey, we can see that 76% of software developers in industry don't know what Green and Sustainable Software is.



94% of companies don't have anyone responsible for Green Software Initiatives.



Most software developers in industry would prefer Awareness, Training, and knowledge of Benefits to help adopt Green and Sustainable Software. This highlights a strong demand for education and resources that makes sustainability practices easier to adopt.

3. Methodology

A mixed methods approach was taken to collect different types of data for analysis. By combining automated bulk scraping of mobile app stores, large-scale GitHub repositories and a public survey on green software adoption, data was collected, analysed and structured into a matrix designed to reflect sustainability challenges and opportunities.

3.1 Scraping

Custom Node.js scrapers were designed to collect metadata from the Google Play Store, Apple App Store, and GitHub to track:

- Low app installs and review counts
- Update frequency
- Signs of program abandonment and inefficiency.

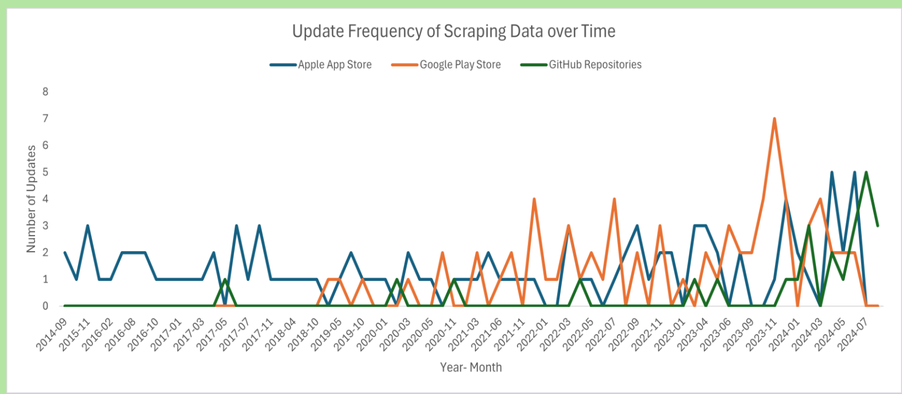
The data collected identified programs that were left to rust in digital space instead of being reused or disposed of sustainably, highlighting missed opportunities where sustainable practices could have extended their lifecycle.

3.2 Survey

An anonymous survey was conducted with ethics approval, targeted at software engineers and developers in industry to gather insights on:

- Awareness of green software engineering
- Adoption in industry
- Challenges faced

Findings showed clear gaps and lack of awareness regarding green software engineering practices and its lack of inclusion in industry.



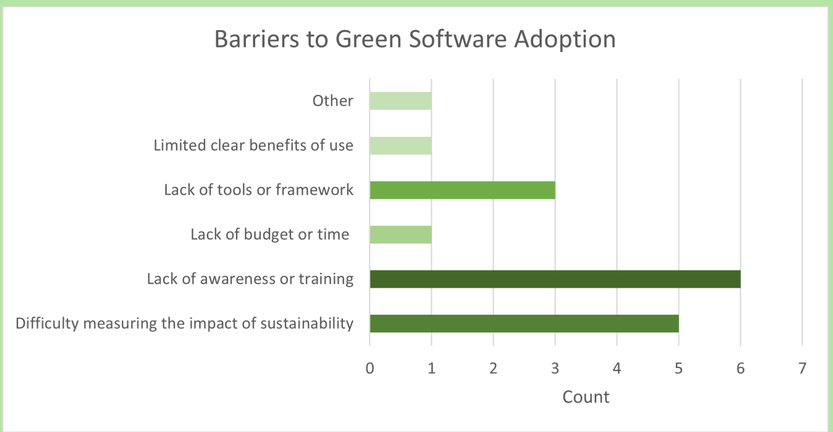
The data suggests irregular update activity across all platforms with many apps and repositories abandoned, deleted or archived, reducing their long term sustainability.

4. Results

Scraped data revealed only a small fraction of apps met potential abandonment criteria defined as less than 1000 installs on Google Play, and fewer than 20 reviews on the Apple app store combined with no updates since before July, 2024. On GitHub, projects inactive since July 2024, highlighted how lifespans could be extended with sustainable practices. Survey findings identified the current state of industry adoption showing that:

- 76% of developers were unfamiliar with Green Software
- 94% of organisations had no dedicated role or team responsible for sustainability.
- Main barriers included lack of awareness, tools, and measurement.

Despite this, there was strong interest in change, with many calling for training, awareness, and clear financial benefits to improve adoption.



The biggest barrier to Green Software Adoption is lack of awareness or training. Many other developers also felt difficulties in measuring the impact sustainability would have. A lesser but also considered barrier was lack of tools or frameworks to consult when incorporating sustainability.

5. Insights Matrix

Aspect	Survey Findings	Scraping Findings	Synthesised Insight
Awareness and Knowledge	76% of developers unfamiliar with Green Software.	Little evidence of green practices in app analysis.	Strong awareness gap. Existing frameworks are not used in industry.
Organisation Adoption	94% of organisations don't have structured support for sustainability inclusion.	Many abandoned apps/repos show no long-term ownership.	Lack of structural investment results in ecosystem abandonment.
Measurement and Tools	A lack of tooling is evident. Developers report no access.	No metrics for sustainability shown in app/repo data.	Absence of tooling reinforces idea that sustainability isn't prioritised.
Value and Motivation	Majority developers rate practices as "slightly valuable."	Low install apps are typically abandoned and wasted.	Sustainability undervalued despite wasted outcomes.
Future Interest	Strong interest for training and learning for adoption.	Patterns of abandonment repeat across platforms.	Strong interest in learning exists but requires early stage adoption and training.